Xijing Wang

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Education

Carnegie Mellon University, School of Computer Science — Pittsburgh, PA
M.S. in Automated Science | Director's Fellowship — *Aug 2025 – May 2027*

Santa Clara University, College of Arts and Sciences — Santa Clara, CA
B.S. in Computer Science (Algorithms), Minor in Economics | REAL Program Scholar — *Sep 2021 – Mar 2025*

Technical Skills

Languages: **Python**, **C++**, Swift, Java, Go, TypeScript/JavaScript, SQL
Backend / Systems: **FastAPI**, REST APIs, SQLAlchemy, PostgreSQL, SQLite, Redis, Docker, Linux/Shell, Git, Pytest
Frontend / Mobile: SwiftUI, UIKit, React Native, React, Redux, **Core ML**, HealthKit
AI Infrastructure: LLM Agents, Agentic Workflows, **RAG**, **coremltools**, **MLX**, **MLC-LLM**, **TVM**, PyTorch

Experience

Apple — Cupertino, CA
*Software Engineer Intern | **Python, Core ML, coremltools, MLX**, Swift* — *May 11 – Aug 21, 2026*
- Built agentic developer workflows that orchestrate cellular ML model training, evaluation, Core ML packaging, and iPhone performance validation into reproducible end-to-end runs.
- Developed software workflow components for the **Call Context** on-device LLM feature announced at **WWDC26**, integrating model artifacts, runtime invocation, and device-side validation.
- Contributed upstream fixes/features to Apple ML tooling: **coremltools** LightGBM conversion [PR #2731] and **MLX** custom Metal kernel export/tracing support [PR #3650].

Open Source Contributor — MLC-LLM & Apache TVM — Remote
*Embedding serving and compiler/runtime support | **Python, C++, TVM*** — *Feb 2026 – Present*
- Contributed to **MLC-LLM** embedding serving/runtime across request scheduling, FFI plumbing, batch prefill, pooling, runtime guards, and Python/C++ parity tests.
- Upstreamed **Apache TVM** dependencies for masked sequence prefill, covering right-padded encoder batches with valid lengths and causal left-padding for decoder-only workloads.

Research Engineer (AI Agent) — Pittsburgh, PA
*Carnegie Mellon University | **Python, FastAPI, Transformers*** — *Jul 2025 – Dec 2025*
- Built a biomedical literature analysis platform with parallel retrieval from **PubMed** and **MedRxiv**, supporting Quick ($<$30s) and Deep ($\leq$3min) structured analysis modes.
- Implemented embedding-based semantic search, reranking, structured report generation, and API workflows for extracting clinical variables and outcomes from research papers.

Machine Learning Research Intern — Santa Clara, CA
*Lab of Cognitive & Computational Neuroscience, Santa Clara University | **Python, TensorFlow**, Shell* — *Jun 2024 – Sep 2024*
- Developed CNN models and data-processing pipelines for neuroscience experiments; automated fMRI preprocessing on an HPC platform with Linux/Shell tooling.

Projects

LocalMelo — Localized AI Agent Framework | ***Python, FastAPI**, LLM Serving* [GitHub] — 2026
- Designed **LocalMelo**, a **local-first** agent framework that separates LLM serving support from the agent runtime, with **planner**, **memory**, **executor**, and **sleep** components for localized AI workflows.

Medical Literature Analyzer | ***Python, FastAPI**, RAG, Transformers, Faiss* [GitHub] — 2025
- Built an AI-powered literature analysis system with retrieval, semantic search, reranking, and report generation for clinical research workflows.

Food Recognition iOS App | *Swift, SwiftUI, **TensorFlow, CoreML**, Python* [GitHub] — 2024
- Built a SwiftUI iOS app using MVVM and a TensorFlow-trained CoreML classifier; achieved 85% accuracy across 100+ categories and optimized inference by 4×.

MasumiRanker — AI Agent Discovery Platform | *Python, FastAPI, SQLite, Faiss, React* [GitHub] — 2025
- Developed semantic agent search with Sentence Transformers and Faiss; built FastAPI/SQLAlchemy services for ratings, recommendations, and review integrity checks.